Eulalia VoterID — Final Milestone Test Report

Reviewed evidence date: May 15, 2026, with additional historical results identified in repository documentation.
System status: technical proof of concept (PoC). This report is not electoral certification, an independent security assessment, or authorization for production.

1. Executive summary

The available evidence supports a technically integrated PoC for selected registration, authentication, SSI, biometric-related, and affiliation flows. The auditable execution baseline in `integration-validation/05-tests` is **Backend: 132 tests, 130 passed and 2 failed**, and **Flutter: 1 test, 1 passed**. The **3/3 frontend** result exists as historical/documented evidence, but it is not considered reproducible in the current state reviewed for this report.

The conclusion is intentionally limited: there is functional and integration evidence for parts of the path, but insufficient evidence to claim comprehensive quality, security, privacy, performance, production operations, or electoral suitability. The two backend failures, low coverage, conditional affiliation validation, absence of a current frontend execution, and lack of a Cardano Preview E2E transaction remain visible.

2. Scope and method

This report consolidates documented results only. No new tests were run to produce the matrix. Each row distinguishes:

- `historical/documented evidence`: a result recorded in a file, log, or README.
- `reproducibility status`: whether the artifact supports repeating the result in the reviewed state.
- `not executed in this report`: no new execution was started for this document.
- `limitations`: the boundary of the inference allowed by the evidence.

The review covered prior `final-report` documents, the five component READMEs, `integration-validation` READMEs, `05-tests` outputs, API/SSI evidence, and the README and suites in `eulalia-backend-cardano`.

Out of scope: the article/publication and closing videos are separate deliverables. They are not evaluated here and are not the requested final delivery index.

3. Reproducibility status

Evidence	Status	Interpretation
.NET backend	`historical/documented evidence`; partial reproducibility	The output contains a final 132/130/2 result, but also records an aborted attempt caused by a missing `Newtonsoft.Json` dependency and configuration warnings. `dotnet test` was not run for this report.
Flutter	`historical/documented evidence`; reproducibility not established	`flutter-test-output.txt` records 1/1. `flutter test` was not run for this report.
Web frontend	`historical/documented evidence`; **not reproducible in the current state**	Bun/Vitest output records 3/3, but the result is historical. `bun run test` was not run for this report.
API/SSI	`historical/documented evidence`; partial reproducibility	JSON responses, logs, and README evidence exist, but there was no new run and no complete reproducible package for all six cases.
Cardano Preview	`not evidenced`	Suites and commands are declared, but there is no `npm test` output and no documented Preview transaction.

4. Final evidence matrix

Dimension	Component	Scenario	Documented method/command	Evidence	Result	Status
Functional accuracy	Backend	.NET test suite	`dotnet test`	`integration-validation/05-tests/dotnet-test-output.txt`, final summary `Total 132; Passed 130; Failed 2`	Two cases failed: `AfiliacionControllerTests.Create_ShouldAddAfiliacion`	**Fail**
Functional accuracy	Backend	Registration, authentication, and domain endpoints	Documented API paths; `dotnet test` as component command	Backend README; logs in `01-api-ssi`	Selected paths have supporting evidence	**Partial**
Functional accuracy	Web frontend	App rendering	`bun run test` / Vitest	`05-tests/bun-test-output.txt`: 1 file, 3 tests passed	Historical 3/3	**Not reproduc**
Functional accuracy	Flutter	Login rendering	`flutter test`	`05-tests/flutter-test-output.txt`: `EulaliaApp renders login page`, `All tests passed`	Historical 1/1	**Pass**
Functional accuracy	SSI API	Province lookup	GET `/api/Provincia`	`integration-validation/01-api-ssi/README.md`	HTTP 200 documented	**Pass**

Dimension	Component	Scenario	Documented method/command	Evidence	Result	Status
Functional accuracy	SSI API	Connection webhook	POST `/api/SSI/webhook`	`01-api-ssi/ssi-webhook-response.json`	HTTP 200; `Webhook processed successfully`	**Pass**
Security/privacy	SSI API	Invitation protection	POST `/api/SSI/invitation/{cedula}` without valid authentication	SSI API README	HTTP 401 documented	**Pass**
Security/privacy	Backend	Affiliation gate	`CredentialIssued` plus biometric enrollment validation	Backend README; biometric failure/log in `01-api-ssi/backend-stdout.txt`	Rule is documented; biometric path failed because relation `biometriaciudadano` was missing	**Partial**
Security/privacy	Cardano backend	PII rejection and SHA-256 commitments	Vitest suites in `tests/validation.test.ts` and `tests/metadata.test.ts`	Test source; no execution output	Validation and privacy claims exist	**Not evidence**
Security/privacy	Identus	SSI health and stack	`curl .../cloud-agent/_system/health`; `docker ps`	`02-identus/cloud-agent-health.json`; `docker-containers.txt`; `health.pdf`; `infra.pdf`	Cloud Agent 1.39.0 and active containers documented	**Pass**
Usability	Web frontend	Login and organization pages	Local server and planned captures	`03-frontend/README.md`; `solicictud.pdf`; `organizaton.pdf`	Login/accessibility documented; captures marked TODO	**Not evidence**
Usability	Flutter	Mobile flow and wallet	`flutter run`; `flutter test`	`04-mobile/README.md`; `screens.pdf`; 1/1 output	One rendering test historically passes	**Partial**
Integration/operational readiness	API + backend + Identus	Invitation, webhook, status, and credential	Documented sequence and SSI APIs	Backend README; `ssi-webhook-response.json`; empty invitation response	SSI 5/5 reported in prior reports; webhook 200 documented	**Partial**
Integration/operational readiness	API integration	Six cases including conditional affiliation	Documented integration tests	Prior reports and root README: 6, 5 passed, 1 conditional	Conditional result occurs when required credential state is unavailable	**Partial**
Integration/operational readiness	Cardano backend	Service and simulated-provider unit/integration suites	`npm test`, `npm run typecheck`, `npm run build`	`package.json`; `tests/config.test.ts`, `service.test.ts`, `blockchain.test.ts`	Suites declared, with mocked blockchain provider in tests	**Not evidence**
Integration/operational readiness	Cardano Preview	Record affiliation/cancellation on chain	Local API + Blockfrost Preview; credentials and funded wallet required	`eulalia-backend-cardano/README.md`	README explicitly states no E2E success claim without valid credentials, funded wallet, and network access	**Not evidence**

5. Metric reconciliation

5.1 Auditable baseline

The baseline adopted by this report is the concrete output in `integration-validation/05-tests`:

Component	Total	Passed	Failed/conditional	Evidence status
.NET backend	132	130	2 failed	`historical/documented evidence`; final result visible in output
Flutter	1	1	0	`historical/documented evidence`; result visible in output
Web frontend	3	3	0	Historical; **not reproducible in the current state**

The `05-tests` README shows a total of 136 (132 + 3 + 1), while the root README and prior reports show 147 after adding API integration (6) and SSI E2E (5): 136 + 6 + 5 = 147. This is not treated as a new execution or a retroactive correction of the output. The figure 147 is retained only as a **historical/documented aggregate**, not as a current report rate.

5.2 Historical documented aggregate

Prior reports document **147 total, 144 passed, and 3 failed or conditional, reported as 98.0%.** That figure is arithmetically consistent when all components are included: backend 132, frontend 3, Flutter 1, API 6, and SSI 5; passed results are 130 + 3 + 1 + 5 + 5 = 144, while the two backend failures plus one conditional validation total 3. The actual scope inconsistency is that the `05-tests` README presents only the 136-component subtotal and does not include API/SSI.

This report therefore retains 147/144/3 as a **historical/documented aggregate**, but not as a new execution or a sufficient approval rate for the milestone or readiness.

5.3 Coverage

Coverage documentation reports approximately **12.4% line coverage** and **25% branch coverage** for the backend. The `tests` README also shows 12.48% lines, 25% branches, and 41.17% methods. These are compatible as line rounding, but they are not system-wide coverage. No comparable executed coverage exists for frontend, Flutter, operational SSI, or Cardano E2E.

6. Milestone acceptance-criteria mapping

Criterion interpreted from documented evidence	Evidence	Assessment
Integrated PoC across backend, web, mobile, and SSI	READMEs, 05-tests outputs, Identus evidence	**Partially and demonstratively met**; does not imply production.
Backend executed and results reported without hiding failures	`dotnet-test-output.txt` 132/130/2	**Met as evidence**, **not met as a fully passing suite**.
Mobile client has at least one tested path	`flutter-test-output.txt` 1/1	**Met within the documented minimum scope**; coverage is limited.
Web frontend verified	`bun-test-output.txt` 3/3	**Historical/documented**, not reproducible in the current state; not a full current acceptance result.
SSI flow documented and connected	webhook 200, health 1.39.0, Docker stack, historical 5/5 reports	**Partially met**; invitation and complete E2E lack a current reproducible package.
Privacy and affiliation controls	JWT 401, credential/biometric gate, Cardano commitment suites	**Partial**; no independent security audit and biometric gate found a missing table.
Operational Cardano Preview integration	Cardano README	**Not evidenced**; only claims/suites and mocks exist, with no real transaction.
Production readiness or electoral certification	No evidence establishes it	**Not met / outside PoC scope**.

7. Risks and limitations

- Two backend tests fail, both associated with the affiliation path.
- Integration logs show `42P01` because relation `biometriaciudadano` does not exist; a port 5219 conflict also appears in a startup attempt.
- Backend coverage is low: approximately 12.4% lines and 25% branches.
- Frontend 3/3 is historical and not reproducible in the current state; there is no current full UI, accessibility, or responsive test.
- Flutter has only one documented test; wallet, camera, biometric, SSI, and device capabilities are not adequately covered.
- The SSI invitation JSON file is empty and planned screenshots remain TODO.
- Identus evidence verifies a documented development environment, not secrets management, availability, recovery, observability, or production deployment.
- The Cardano tests use mocks and do not test a real Preview transaction. No documented credentials, funds, or network access are available.
- Backend logs indicate sensitive-data logging enabled, suitable only for development and risky in a shared environment.
- The PoC does not demonstrate legal compliance, sufficient privacy, institutional governance, independent security, scalability, fraud resistance, or electoral certification.

8. Recommendations

1. Fix the two backend failures and the `biometriaciudadano` schema mismatch; add regression tests for affiliation gates.
2. Re-run backend, frontend, and Flutter from clean environments and preserve complete outputs, versions, dependencies, and execution date.
3. Increase backend coverage and add explicit security, privacy, error, performance, and recovery tests.
4. Restore reproducible frontend execution and cover authentication, dashboard, empty states, errors, accessibility, and responsive behavior.
5. Expand Flutter coverage to wallet, permissions, biometrics, SSI, offline/error handling, and a device matrix.
6. Separate mocked unit/integration tests from a real Cardano Preview test; run the latter only with protected secrets, a funded wallet, and transaction-hash evidence.
7. Disable sensitive-data logging, replace development secrets, and define retention, access, revocation, and recovery controls before any pilot.
8. Conduct independent security, privacy, governance, legal, and electoral-requirements reviews before presenting the system as a pilot candidate.

9. Conclusion

The evidence supports closing a technical PoC validation for selected paths, not production acceptance. The honest result is: **backend 130/132 passed with 2 failures; Flutter historical 1/1; frontend historical 3/3 and currently not reproducible; SSI/API partially documented; Cardano Preview E2E not evidenced; backend coverage low**. These failures and absences are part of the result and must remain visible in any later delivery.

10. References

- Integration evidence
- Test baseline outputs
- Backend output
- Historical frontend output
- Flutter output
- SSI API evidence

- Identus evidence
- Frontend evidence
- Mobile evidence
- Backend
- Frontend
- Mobile application
- Atala PRISM / Identus integration
- Cardano backend

External references

- Public Digital Voter ID using Atala PRISM technical documentation
- External article by Extra
- Vimeo closing video (English)
- Vimeo closing video (Spanish)

Note: This document does not create or modify the final delivery index.

Direct URLs

https://www.extra.ec/noticia/tecnologia/identidad-digital-tecnologia-evitar-afiliaciones-politicas-fraudulentas-166630.html
https://vimeo.com/1239350128
https://vimeo.com/1239351564